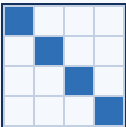


Four formulas, one operation

All four compute the same thing — the closest point of the target to **b**. They differ only in what is assumed about the basis, and the formula gets simpler as the assumption gets stronger.

ONTO A SINGLE VECTOR

1



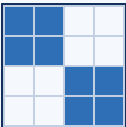
A line | the target is $\text{Span}\{\mathbf{a}\}$

FORMULA	REQUIRES	COST PER PROJECTION
$\frac{\mathbf{b} \cdot \mathbf{a}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}$	$\mathbf{a} \neq \mathbf{0}$	two dot products

The base case, and the one every formula below is built from. The denominator is what normalisation removes — for a unit vector it is 1 and disappears.

ONTO A SUBSPACE — BASIS ASSUMED ORTHOGONAL

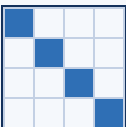
2



Orthogonal basis | $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$

FORMULA	REQUIRES	COST PER PROJECTION
$\sum \frac{\mathbf{b} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i} \mathbf{u}_i$	pairwise orthogonal, nonzero	$2k$ dot products

A sum of independent line projections. Orthogonality is what makes the terms independent — each coefficient can be computed without knowing the others, and adding a basis vector does not disturb them.



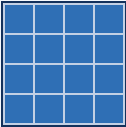
Orthonormal basis | orthogonal and $\|\mathbf{u}_i\| = 1$

FORMULA	REQUIRES	COST PER PROJECTION
$\sum (\mathbf{b} \cdot \mathbf{u}_i) \mathbf{u}_i$	pairwise orthogonal, unit length	k dot products

The denominators vanish because each is 1. This is the form worth arranging for — see **orthonormal operations**, where every operation simplifies the same way.

ONTO A SUBSPACE — NOTHING ASSUMED

1



Arbitrary basis | columns of A span the target

FORMULA	REQUIRES	COST PER PROJECTION
$A(A^T A)^{-1} A^T \mathbf{b}$	columns of A independent	a $k \times k$ inverse

The general case, and the expensive one. $(A^T A)^{-1}$ exists exactly when the columns are independent — and when they happen to be orthonormal, $A^T A = I$ and this collapses to the row above.

Reading upward is the argument for **Gram–Schmidt**. The general formula needs a matrix inverse; orthogonality removes the coupling between coefficients; normalisation removes the denominators. Each assumption bought by preprocessing is a computation avoided on every projection afterwards.